

XRPL Validator — Daily Engineering Report • 2026-08-03

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 6 commits (xrpl-validator 6)

Engineering changes

- **fix(ledger): the POOL can activate a strand, and an active strand reaps** (6e631e7 , xrpl-validator) rippled filters strands on `qualityUpperBound` — an OPTIMISTIC bound on what the strand could conceivably do (StrandFlow.h:696-699, `qualityUpperBound(sb, *strand) < *limitQuality => continue`). For a book with an AMM that bound includes the pool. A strand whose every BOOK level is worse than the limit therefore still gets built, its offer stream still steps, and the dead offers it steps past are still reaped — even ...
- **feat(ledger): reverse pass — size each payment hop to what the next needs** (9ac6bcc , xrpl-validator) rippled flows a strand in TWO passes: a REVERSE pass walking back from the requested output working out how much input each step needs (StrandFlow.h:105-230, `for (auto i = s; i--;) strand[i]->rev(...)`), re-executing a limiting step when one binds), then a FORWARD pass bounded by what is actually available. Ours was forward-only, so an intermediate hop sold its WHOLE carry rather than buying what the hop after it ...
- **feat(ffl): capture libxrpl's decision trace under XRPL_FFI_TRACE** (8b5a197 , xrpl-validator) The shim already had a CapturingSink wired into `xrpl_apply_with_mutations` and surfaced through `xrpl_result_last_fatal` — but gated at `Severity::Fatal`, so everything rippled narrates was thrown away. Lift it to `Trace` when `XRPL_FFI_TRACE` is set, and dump it per transaction with `XRPL_FFI_TRACE_TX=`. Both OFF by default: gate runs stay byte-identical and trace capture is neither free nor small.
- **fix(ledger): round a fill's XRP input by which side bounds it** (d6f7589 , xrpl-validator) The taker's input was always ceiled to whole drops. rippled's direction depends on which side of the fill is fixed — its rev/fwd split: - the taker's remaining WANT bounds it: the input is computed FROM that output (`rev`) and rounds UP, so the taker pays enough; - the maker's FUNDING bounds it: the output is what is fixed and the input follows it (`fwd`), rounding DOWN.
- **fix(ledger): judge a fill's achieved quality exactly, as rippled does** (133fd89 , xrpl-validator) rippled discards a pass outright when the quality it actually realised misses the limit: `if (limitQuality && q < limitQuality && (!adjustedRemOut || !withinRelativeDistance(q, limitQuality, Number(1, -7))))`
- **fix(ledger): one AMM turn per pass, and a rejected pass ends the crossing** (8152aca , xrpl-validator) Three pieces, each only visible once the previous one landed, all read off libxrpl's own trace rather than inferred.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **DID-Linked On-Ledger Reserve Attestations** (audit) Add a new owner-reserved ltRESERVE_ATTESTATION ledger object at keylet::reserveAttestation(issuer, assetID), creatable/updatable only via a new Reserv
- **Confidential MPT Balances, Issuer-Clawback Preserved** (privacy) Add an lsfMPTConfidential flag to MPTokenIssuance and replace the plain integer balance on MPToken with a Pedersen commitment sfBalanceCommitment when
- **XChain Bridge Settlement Finality Certificates** (interop) Add a new protocol-minted (no owner reserve) ltBRIDGE_FINALITY_CERT ledger object at keylet::bridgeFinalityCert(bridgeID, xChainClaimID), created insi

Appendix — commits

Time	Repo	Commit	Subject
01:26	xrpl-validator	6e631e7	fix(ledger): the POOL can activate a strand, and an active strand reaps
01:27	xrpl-validator	9ac6bcc	feat(ledger): reverse pass — size each payment hop to what the next needs
10:21	xrpl-validator	8b5a197	feat(ffi): capture libxrpl's decision trace under XRPL_FFI_TRACE
11:22	xrpl-validator	d6f7589	fix(ledger): round a fill's XRP input by which side bounds it
12:09	xrpl-validator	133fd89	fix(ledger): judge a fill's achieved quality exactly, as rippled does
13:11	xrpl-validator	8152aca	fix(ledger): one AMM turn per pass, and a rejected pass ends the crossing